

Full Length Article

Photopolymerized biomimetic self-adhesive Polydimethylsiloxane-based amphiphilic cross-linked coating for anti-biofouling

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ABSTRACT

Biofouling as a global problem has made detrimental impact on many fields, especially for marine industry. Polydimethylsiloxane-based coatings are environment-friendly and exhibiting good fouling release performance in the marine coating application. However, their intrinsic low surface energy and high hydrophobic character lead to poor fouling resistance property and weak adhesion to substrate. In this work, we firstly fabricate self-adhesive PDMS-based amphiphilic cross-linked coatings with good anti-biofouling performance via synthesizing dopamine methacrylamide which then copolymerized with mercaptopropyl methylsiloxane- dimethylsiloxane, poly (ethylene glycol) diacrylate under UV photo-cross-linking. The coating exhibits devisable heterogeneous nanoscale mosaic chemical surfaces caused by micro phase segregation, effectively resistance to nonspecific protein adsorption and marine fouling organism (*P. tricornutum*) attachment which are mainly ascribed to the unique surface topography and chemical heterogeneity. Moreover, the resulting coatings exhibit significantly improved adhesion to substrates, due to complex interfacial catechol chemistry reaction. This simple design provides a novel methodology for preparing an efficient and eco-friendly anti-biofouling PDMS-based amphiphilic coating.

1. Introduction

Today, polydimethylsiloxane (PDMS) based materials have been a new candidate against marine biofouling caused by unwanted adsorption of proteins, cells, bacteria and microorganisms [1]. Owing to their low surface energy and low mechanical modulus, the PDMS-based coatings demonstrate excellent fouling release performance leading to easy removal of biofoulants attached on the ship hulls or other subsea equipment through surface cleaning methods [2,3]. More importantly, the nontoxic PDMS-based coatings have little impact on marine ecosystem compared with the traditional biocidal coatings [4,5].

Generally, marine anti-biofouling coatings are expected to possess both fouling release and fouling resistance properties so as to effectively combat with various marine biofoulant fouling [4,6,7]. Nevertheless, PDMS materials are easy to fouling by contaminants due to the intrinsic high hydrophobic performance [8,9]. It has been reported that PDMS-based coatings on marine equipment or vessels are vulnerable against nonspecific protein, cell, bacteria or microorganisms, which provides initial fouling layer for further colonization of more complicated organisms, resulting in final overall biofouling [10,11]. Therefore, it is extremely necessary to improve the fouling resistance performance of PDMS coatings.

The amphiphilic surfaces have been found to effectively inhibit the settlement of biofoulants due to their micro and nanoscale surface topographies, which decrease interfacial bonding with macromolecular contaminants [10,12–15]. Thus, fabricating amphiphilic system via introducing hydrophilic components into PDMS is regarded as one promising method to enhance its antifouling performance [11,16]. Polyethyleneglycol (PEG) [17–19], Poly (ethylene oxide) (PEO) and zwitterionic polymers [20–22] with high hydrophilicity have been found to facilitate generate surface with excellent antifouling property. The immobilization of PEG into PDMS as a convenient way has been widely used in recent years, the PEG component is able to form a bound water layer at the interface, which contributes to inhibiting the adsorption of proteins as well as the adhesion of microbe [18,23]. Moreover, the electrically neutral character of PEG is also helpful to resist various protein adsorption [24].

As we all know, the indispensable adhesion to substrate is necessary for marine coatings. However, the PDMS coatings reveal poor adhesion on many substrates due to their intrinsic low surface energy character [25,26]. Many strategies, such as fillers blending, chemical modification, have been adopted to improve the adhesion [27–29]. Recently, 3,4-dihydroxyphenylalanine (DOPA) in mussel adhesive protein has

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been widely studied due to its impressive adhesion properties [30,31]. Lots of low fouling coatings have been prepared via treating catechol groups as a biomimetic anchor, including hydrophilic DOPA-PEG [19,32], polycarboxybetaine methacrylate (DOPA-PCB) [33,34], polysulfobetaine methacrylate (DOPA-PSB) [32] or amphiphilic catechol-containing fluorinated copolymers [35], even hydrophobic fluoropolymer [36]. All the coatings exhibit long-term stability on account of the presence of catechol groups and expected anti-fouling performance. Although catechol-containing amphiphilic coatings have been studied for antifouling surfaces, the introduction of catechol derivatives into PDMS moieties for enhanced adhesive performance via polymerization has not been explored so far. Usually, thermal free radical copolymerization is adopted to introduce catechol moieties into polymeric structures, which is more limited and inefficient. Several reports have confirmed the successful preparation of polymer cross-linked structures containing catechol functionality via photopolymerization [37–39].

Herein, an easy one-step method is employed to prepare novel biomimetic PDMS-based amphiphilic coating, where dopamine methacrylamide firstly copolymerized with mercaptopropyl methylsiloxane- dimethylsiloxane, poly (ethylene glycol) diacrylate via UV photo-cross-linking. The resulting amphiphilic PDMS-based coatings demonstrate a combination of excellent properties, i.e., excellent adhesion on various substrates, low swelling ratio in water, excellent anti-biofouling performance and high thermal stability, which is a suitable candidate for marine anti-fouling coating application.

2. Materials and methods

2.1. Materials

Dopamine hydrochloride was purchased from Alfa Aesar Chemical Co., Ltd. Methacrylic anhydride (94%), Fluorescein isothiocyanate and photoinitiator Irgacure 651 (DMPA) were purchased from Aladdin Industrial Corporation. Poly (ethylene glycol) diacrylate (PEGDA; 600 g/mol, Aldrich) was purified through a basic alumina column to remove the radical inhibitor. (Mercaptopropyl) methylsiloxane- dimethylsiloxane (PDMS-SH) ($M_n = 5500 \text{ g}\cdot\text{mol}^{-1}$, PDI = 1.40, mole [(Mercaptopropyl) methylsiloxane] % = 100,) was purchased from Gelest and used without any purification. Bovine serum albumin (BSA, $67000 \text{ g}\cdot\text{mol}^{-1}$) was purchased from Energy Chemical, and used as received. All other reagents and solvents were purchased from Sinopharm Chemical Reagent (Shanghai, China) and used without further purification. Phosphate Buffered Saline (PBS; pH = 7.4, 10 mM, 1.76 mM KH_2PO_4 , 10.14 mM Na_2HPO_4 , 137 mM NaCl, 2.7 mM KCl) was self-prepared in lab.

2.2. Synthesis of dopamine methacrylamide (DMA)

DMA was synthesized according to method described in the literature [40]. Firstly, sodium carbonate (2.0 g) and sodium tetraborate (9.5 g) were dissolved in 100 mL of doubly distilled water and degassed with nitrogen for 30 min. Then, Dopamine hydrochloride (2.5 g) was added into the mixture, following the dropwise addition of methacrylic anhydride (3.0 mL) in 10 mL THF. The NaOH solution was then added into the solution to keep the reaction in alkaline environment. Finally, the solution was stirred for an additional 24 h at 25 °C under bubbling nitrogen. After that, about 50 mL of ethyl acetate was added into the mixture and then filtrated. The pH of the filtrate was acidified to less than 2 using 6 M HCl and extracted with 50 mL of ethyl acetate for three times. The extracted clear brown organic layer in the ethyl acetate was dried over MgSO_4 and then purified by column chromatography. After the solution was concentrated with a rotary evaporator, the mixture was added to 250 mL ice hexane under vigorous stirring to precipitate. To purify, the resulting light brown solid was dissolved in 20 mL of ethyl acetate and precipitated in hexane. Finally, DMA was obtained as

a gray solid after dried in a vacuum overnight (50 °C). ^1H NMR (600 MHz, $\text{DMSO}-d_6$, δ): 8.75 (OH, H), 8.63 (OH, H), 7.93 (NH, 1H), 6.63 (Ph, 1H), 6.57 (Ph, 1H), 6.44 (Ph, 1H), 5.62 (HC=C, 1H), 5.30 (HC=C, 1H), 3.21 (NH- CH_2 -, 2H), 2.57 (CH_2 -Ph, 2H), and 1.84 (CH_3 , 3H).

2.3. Preparation of biomimetic self-adhesive PDMS-based amphiphilic Cross-Linked coatings

(Mercaptopropyl) methylsiloxane- dimethylsiloxane (S), poly (ethylene glycol) diacrylate (P), dopamine methacrylamide (D), DMPA were firstly dissolved in DMF until a clear and homogeneous mixture. After the mixture was clarified by syringe filter (PTFE, 0.45 μm , 25 mm), the coatings were prepared by casting the mixtures on glass slides, steel or Al substrates and then carried out on UV Curing System (UVATA, OEM, $\lambda = 365 \text{ nm}$) under $1.3 \text{ W}\cdot\text{cm}^{-2}$ light intensity for 150 s. And The UV light was at approximately 10 cm distance from the coating. Finally, the coating samples were soaked in DI water to remove residual DMF and dried in a vacuum (65 °C, overnight). The thickness of the coatings was about 25 μm .

3. Analysis and characterization

3.1. ^1H NMR characterization

^1H NMR spectra was obtained using an AVANCE 600 MHz spectrometer (Bruker) with tetramethylsilane as an internal standard. DMA was dissolved in $\text{DMSO}-d_6$ at room temperature.

3.2. FTIR spectra

Fourier transforms infrared spectra were obtained using a Nicolet Instrument Nicolet 8700 spectrometer in attenuated total reflection. Spectra were recorded with a 4 cm^{-1} resolution and 32 scans a measurement at a wave length range of 500–4000 cm^{-1} .

3.3. Water swelling tests

The coatings on glass slides were soaked in deionized water for different time. The weight percentage of swelling degree was determined by the equation

$$W_{\text{swelling}}\% = \frac{m_2 - m_1}{m_1} \times 100\% \quad (1)$$

The m_1 and m_2 are the mass of the dry samples and wet samples, respectively.

3.4. Thermogravimetric analysis

TGA was carried out by NETZSCH TG209F1. Samples (no more than 5.0 mg) in alumina crucible were heated from room temperature to 800 °C with a heating rate of 10 K min^{-1} under a nitrogen atmosphere (flow rate: 40 mL min^{-1}).

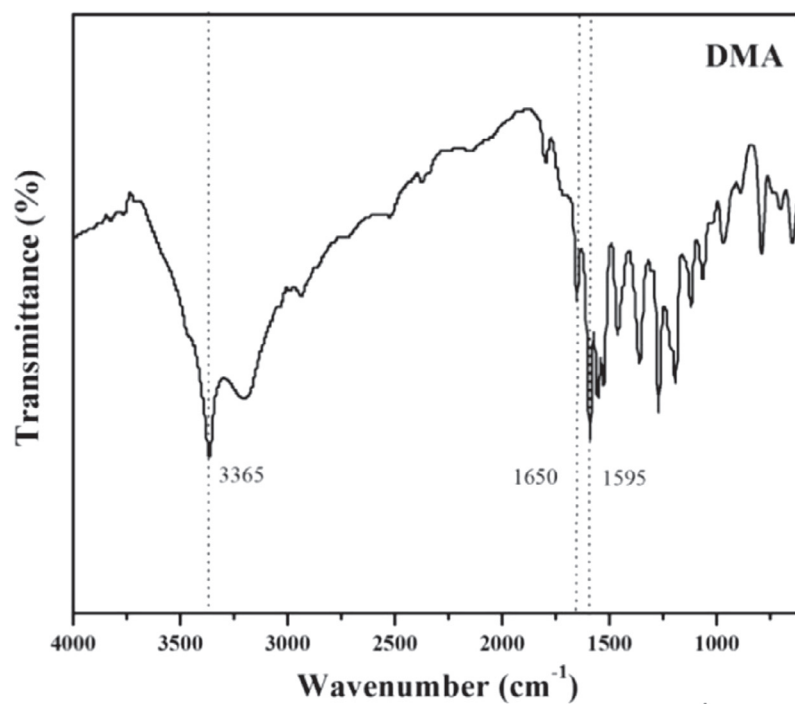
3.5. X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy was performed on a Thermal Scientific K-Alpha spectrometer equipped with a monochromated Al K X-ray source (Al K α , 1486.6 eV). Survey and high-resolution spectra were obtained at takeoff angles ($\theta = 90^\circ$) with respect to the sample surface.

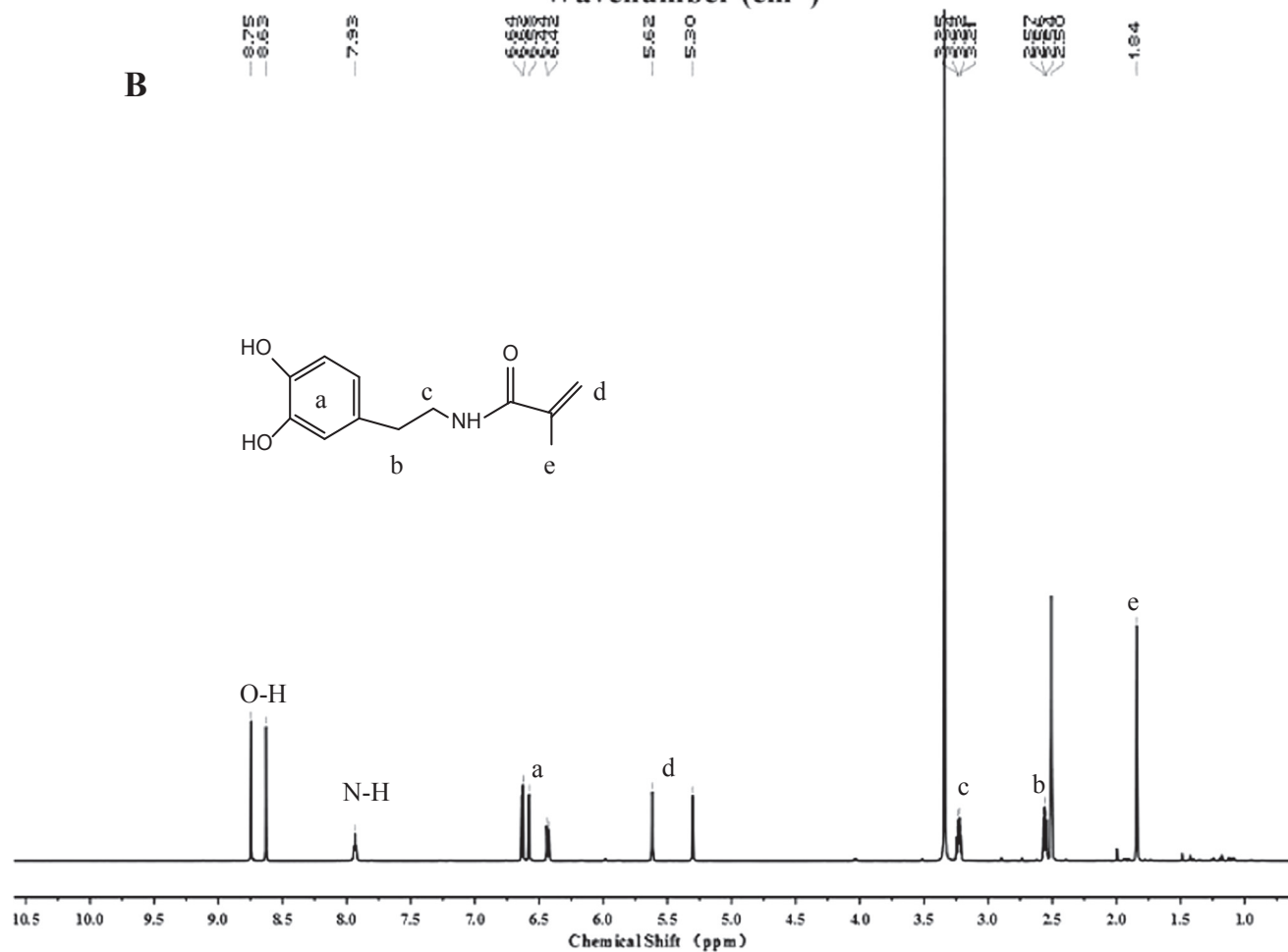
3.6. Water contact angles

Water contact angles (WCA) of the coatings were measured using a telescopic goniometer (OCA40, Dataphysics, Germany) by the sessile

A



B



Scheme 1. The preparation of biomimetic amphiphilic PDMS-based cross-linked antifouling coatings.

drop (3 μ L) method at room temperature. The surface contact angles were taken five spots for each sample, and the mean value of the three determinations.

3.7. Atomic force microscopy

The surface morphology of coatings was acquired by using Pico Image software (Agilent 5500). The Sq roughness values of the coatings were investigated with a silicon tip (160 μ m, 165–175 KHz) with a nominal spring constant of 40 N/m in the tapping mode under ambient conditions.

3.8. Adhesion test

In order to evaluate the adhesion performance on various substrates, two different methods were used in our research. According to ASTM D-3359, the crosshatch adhesion test was carried out via scratching into the coating surface and then tape was adopted to remove the coating from the grid. The adhesive properties were assessed via the loss of the film by the tape, which was graded on a scale of 5B to 0B. The class 5B meant the lattice scratched on films was barely damaged and the cut edges remained smooth, class 4B meant less than 5% area was removed from the lattice, and class 1B was described that 35–65% area of the film was removed. If the damaged region ratio was bigger than 65% of the film, it was classified as 0B. The Pull-off adhesion test was carried out using PosiTest AT-A automatic adhesion tester from DeFelsko according to ASTM D-4541. An aluminum dolly with a diameter of 20 mm was adhered to the surface of the coating on steel substrate. Pull-test adhesion data was obtained by detaching the dolly from the substrate at the speed of 0.2 MPa·s⁻¹. Five different regions for each sample were tested to obtain an average value.

3.9. Protein adsorption test

To intuitively determine fouling properties of coatings, the samples were immersed in 50 μ g·mL⁻¹ FITC-labeled BSA (prepared in self-laboratory) for 24 h at room temperature in darkness, and then cleaned by PBS solution and dried with nitrogen. The FITC-labeled BSA adsorption on the samples was observed with fluorescence microscopy (BX-51, Japan Olympus) excited by an argon ion laser ranging from 488 to 638 nm.

To quantitatively test the protein adsorption, all samples (1 cm $\times$ 1 cm) were pre-wetted in PBS (pH = 7.4) buffer solution, followed by BSA or lysozyme solution incubating for 12 h at room temperature. After being rinsed 5 times with PBS solution, the samples were placed into 1 mL aqueous solution (including 1 wt% sodium dodecyl sulfate (SDS)) and then vibrated in a shaking bath for 2 h at room temperature to remove the protein adsorbed on the surface. And the protein elution solution of each sample was added in a 96-well plate containing BCA reagent (Pierce) at 37 °C for 30 min. The amount of the adsorbed protein was determined by the adsorption intensity at 562 nm using microplate reader (Multiskan FC, Thermo, America). The calibration curve was pre-set by measuring different protein concentrations in SDS solution. The result data was the mean value of five samples for each sample. The control is commercial PDMS (sylgard 184) coating.

3.10. Biofouling assays

The *P. tricornutum* was cultured in a climate chamber at 16 °C with a light intensity of 3000Lx. Then 10 mL diatom suspension was added to petri dish containing coated glass slide. After a period of 7 days at 16 °C in the climate chamber, slides were washed slightly with artificial sea water to remove diatoms that not attached to the surfaces. The attached diatoms on the coating surfaces were characterized by optical microscope imaging system and the density of cells was determined using a hemocytometer.

Table 1

Different mass composition of coating casting solution.

Samples	PDMS-SH (g)	PEGDA (g)	PEG ratio (w%)	DMA(g)	DMA ratio (w %)	DMF (g)
S-P-0	0.50	1.09	68.6	0.00	0	14.30
S-P-D-1	0.50	0.82	53.3	0.20	13.2	13.68
S-P-D-2	0.50	0.65	44.8	0.32	21.7	13.23
S-P-D-3	0.50	0.36	26.2	0.54	38.2	12.60
S-P-D-4	0.50	0.27	19.6	0.62	45.2	12.50

4. Results and discussion

4.1. Synthesis

Scheme 1 exhibits the synthetic strategy of the biomimetic amphiphilic PDMS-based cross-linked antifouling coatings via thiol-ene click photopolymerization reaction. The synthesized DMA (D), and PDMS-SH (S), PEGDA (P) were dissolved in DMF with the presence of 1.5 wt% DMPA photoinitiator for crosslinking process. The reactive allyl groups of DMA and PEGDA undergo the cross-linking with PDMS via click chemistry to give cross-linked networks. In the precursor solution, the weight of PDMS-SH was kept constant and the molar ratio of alkene and thiol functional groups was maintained 1:1 in order to acquire completely crosslinking. Finally the resulting coatings were soaked in DI water to remove residual DMF and dried in a vacuum. In addition, the polymer mixture content was about 10 wt% in DMF solution. The different compositions of the sample coatings were summarized in **Table 1**. The S-P-0 coating was prepared only for the adhesion test not for antifouling characterization due to its fragile character.

The DMA monomer synthesized is characterized by FT-IR and ¹H NMR spectra. As shown in **Fig. 1A**, the absorption bands in FT-IR at 3365 and 3203 cm⁻¹ are attributed to amino (–NH) and phenolic hydroxyl groups (–OH) stretching vibration. The peaks at 1650 cm⁻¹ and 1595 cm⁻¹ correspond to the stretching of amide (–N–C=O) and C=C, respectively. The absorption bands of 1358 cm⁻¹, 1193 cm⁻¹ and 1270 cm⁻¹ are assigned to the benzene ring. The ¹H NMR spectra (**Fig. 1B**) shows the signals at δ 6.63, 6.57, 6.44 ppm and δ 8.75, 8.63 ppm are assigned to the aromatic and hydroxyl groups protons of the catechol group, respectively. The chemical shift at δ 1.84 ppm is assigned to the protons of methyl, while δ 5.30 and 5.62 ppm correspond to the protons of allyl group. As a result, ¹H NMR and FT-IR demonstrate a successful preparation of DMA monomer.

Fig. 2A shows the ATR-FTIR spectra of PDMS-SH, PEGDA, S-P-D-1, which is able to directly determine the chemical structures of the amphiphilic PDMS-based coatings. The peak at 2556 and 1626 cm⁻¹ are assigned to the thiol group and typical carbon double bond in PDMS-SH and PEGDA, respectively, which disappear after photocrosslinking in S-P-D-1 coating. The result confirms the successful and completely thiol-ene click polymerization reaction. **Fig. 2B** shows the ATR-FTIR of coatings with different DMA and PEG content. The typical peak at 1650 cm⁻¹ and 1730 cm⁻¹ in S-P-D-1 correspond to the stretching of amide (–N–C=O) and O–C=O from DMA and PEG segments, respectively, and the absorbance signal at around 3286 cm⁻¹ is assigned to the phenolic hydroxyl in catechol group. All the results indicate that the biomimetic amphiphilic PDMS-based coatings have been prepared via photopolymerization.

In our research, the catechol groups are introduced into our coatings to endow enhanced self-adhesive property. The catechol groups enable to form reversible non-covalent or irreversible covalent interactions to bind on various surfaces. UV–Vis spectrum is used to monitor the structure changes of catechol groups during photopolymerization. As shown in **Fig. 2C**, before photo curing, the characteristic peak at 283 nm is assigned to the catechol group in DMA which is in accordance with [41]. While after exposing to UV for 150 s, the peak intensity at 330 nm is increased, which is ascribed to the existence of quinone structures,

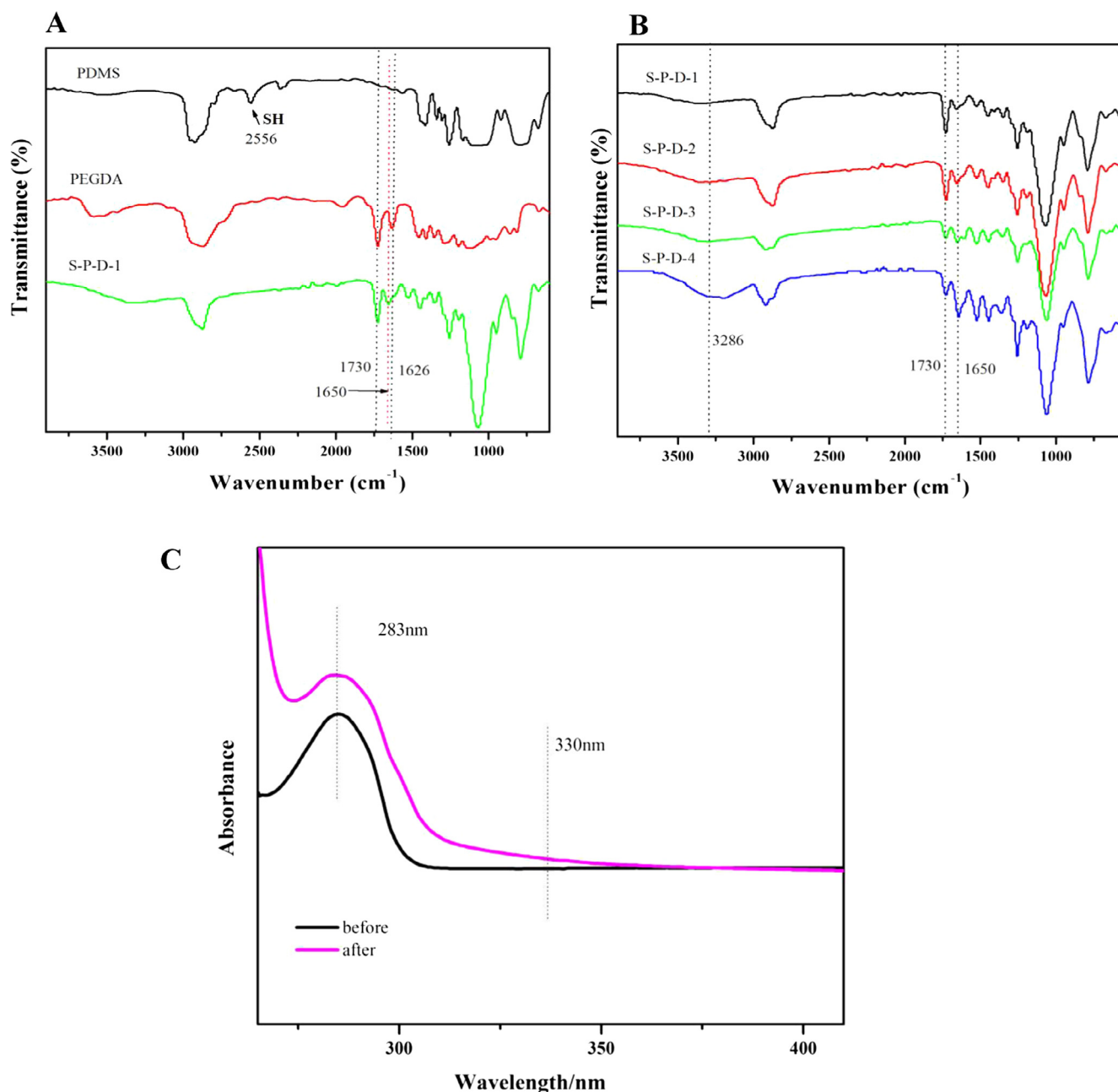


Fig. 1. FT-IR spectra(A) and ¹H NMR spectra(B) of dopamine methacrylamide (DMA).

indicating that part of catechol groups are oxidized to form semi-quinone and quinone [36]. It has been demonstrated that quinone tends to form dimers with another catechol moiety or react with nucleophiles, resulting in interfacial covalent crosslinking [37]. Thus, the quinone groups during photo-curing possibly contribute to enhance the adhesive of coatings.

4.2. Swelling test

It is necessary for anti-biofouling coatings to assess their swelling degree in water considering their application in marine industry or medical implants[39]. As shown in Table 2, the swelling degree in the same immersion time is gradually decreased with increasing DMA content, and the S-P-D-4 shows the least swelling degree in all the tested coatings. Although the swelling degrees of the samples are increased during the immersion time reaches to 160 h, the highest swelling degree is only 3.25%. All in all, the coatings exhibit very low

swelling degree after 7 days of water immersion, indicating their good resistance to the aquatic environment.

4.3. Thermal properties

Thermal analysis is used to study the effect of changes in chemical composition on the thermal stability of the amphiphilic PDMS-based coatings. Fig. 3 shows that thermal destruction temperature (T_d) values of these coatings are increased with increasing DMA content in coatings. Obviously, the existence of DMA monomer is the key to affect the thermal degradation behavior of the coatings. The coating S-P-D-4 with the highest content of the DMA moieties exhibits the highest T_d value while S-P-0 shows the lowest T_d value. The introduction of catechol groups provide rigid structure such as benzene ring and dihydroxyphenyl group, which leads to increased T_d to polymer films. Moreover, the existed catechol groups after curing tend to generate strong hydrogen-bond or other multivalent interaction.

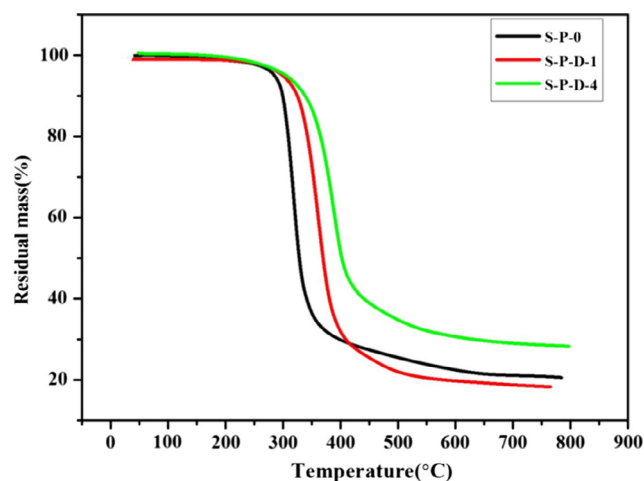


Fig. 2. (A) ATR- FTIR spectra of PDMS, PEGDA and S-P-D-1. (B) ATR- FTIR spectra of prepared coatings. (C) UV-Vis spectra of mixture before and after UV light curing.

Table 2

Swelling degree of PDMS-based amphiphilic coatings in water for different time.

Samples	Swelling degree ^a (wt %)			
	24 h	36 h	72 h	168 h
S-P-D-1	1.30 ± 0.04	1.45 ± 0.05	2.84 ± 0.06	3.25 ± 0.10
S-P-D-2	0.95 ± 0.10	1.25 ± 0.05	2.78 ± 0.20	2.80 ± 0.10
S-P-D-3	0.53 ± 0.05	0.64 ± 0.02	0.96 ± 0.05	1.25 ± 0.10
S-P-D-4	0.20 ± 0.05	0.25 ± 0.05	0.52 ± 0.10	0.90 ± 0.20

^a Calculated by Eq. (1).

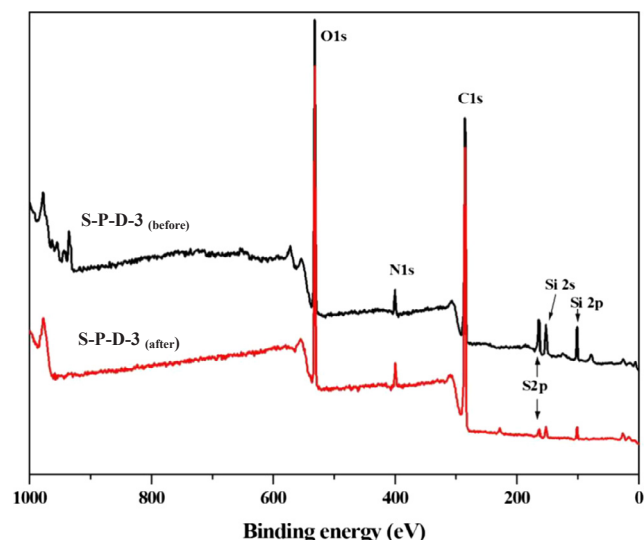


Fig. 3. The typical TGA curves of PDMS-based amphiphilic coatings.

4.4. Surface chemical composition and surface topography

XPS is adopted to investigate the change of surface chemical composition of the coatings before and after being immersed in water for 7 days. As shown in Fig. 4, the XPS result reveals that the S-P-D-3 coating surface possesses all major elements from three components. As shown in Table 3, there is a large increase in carbon content and decrease in silicon content, and a slightly increase in oxygen content on S-P-D-3_(after) coating surface. This phenomenon is attributed to a surface reconstruction. When the coating is immersed in water, the PEG chains

tend to migrate to coating surface due to its hydrophilic character while the PDMS chains go away from the surface. The results clarify that abundant PEG chains are enriched in coating surface in water environment.

Atomic force microscopy (AFM) is used to investigate the amphiphilic surface topography of the PDMS-based amphiphilic coatings. It is seen clearly that the 3D phase images reveal a formation of highly ordered domains on nanoscale. Importantly, the relative content of the three components have large impact on the dimensions of the phase-segregated domains. As shown in Fig. 5(a–c), the surface roughness (Sq) is increased from 4.54 nm to 13.8 nm with increasing DMA content, and all the surfaces indicate regular domains with island-like structure. Interestingly, when the DMA content is further increased to higher ratio, as shown in Fig. 5(d), the average size of distributed domains is larger and the roughness is reduced to 10.1 nm. The dramatically morphologies of these coatings may be attributed to two primary reasons. Firstly, the hydrophobic PDMS and hydrophilic PEG segments tend to phase segregation due to their thermodynamically mutual incompatibility. Secondly, the dopamine group (DMA) plays an indispensable role in the formation of coating surface morphology. It has been demonstrated that polymers containing abundant catechol are likely to aggregate because of the amphiphilic nature of catechol group [42]. Moreover, the catechol group tends to generate complex interface interactions including covalent bonds, hydrogen bonds and crosslinking reaction under photopolymerization [37,43]. Hence, appropriate DMA content in coating contributes to improve surface roughness but excessive content leads to aggregation. And the protrusions on coating surface are thought to be hydrophilic domains. This combination of complex nanoscopic topography and amphiphilicity may present obstacles for settling organism, resulting in passive resistance to biofouling.

4.5. Surface wettability

The hydrophilicity of PDMS-based amphiphilic coating is significant for resisting biofouling. In order to evaluate the wettability of the cross-linked coatings, the water contact angle measurement is adopted. As expected, the static water contact angle decreases from 77° to 56° with increasing amounts of PEG content, as shown from coating S-P-P-4 to S-P-D-1 in Fig. 6. Moreover, the WCA of S-P-D-1 coating is rapidly decreased from the initial 56° to 26° in 30 s, and that of the S-P-D-4 coating changes from 77° to 64°. The reason for significant decreased WCA of S-P-D-1 coating is likely due to the PEG chains reorganization, when the interface is changed from air/polymer to water/polymer. There is no doubt that all the coatings exhibit hydrophilic character compared with the PDMS coating after modified with PEG and DMA.

4.6. Adhesion test

The good adhesion to the substrate is necessary for anti-biofouling coatings applications [25]. In our research, Cross-hatch and Pull-off adhesion tests were used to investigate the adhesive properties of the amphiphilic PDMS-based coatings on various substrates. As shown in Table 4, the S-P-0 without DMA moieties reveals terrible adhesion performance i. e. 0B. With increasing content of DMA, the adhesive property of S-P-D-4 sample is improved significantly for all the glass, aluminum and steel substrates, where the adhesion class is improved to above 4B, due to the strong hydrogen bonding, π - π aromatic interactions, or metal-ligand complexation generated by catechol chemistry reactions on different interfaces. Moreover, the adhesion strength to steel substrate is quantitatively measured via Pull-off test. As shown in Fig. 7, the adhesion strength of S-P-0 coating without DMA is 0.52 Mpa. In contrast, other coatings are increased from 1.21 Mpa to 2.52 Mpa with increased DMA content, which is high enough for their application in marine environment for adhesion strength above 1.0 MPa [25]. Definitely, the presence of biomimetic DMA segments play important role

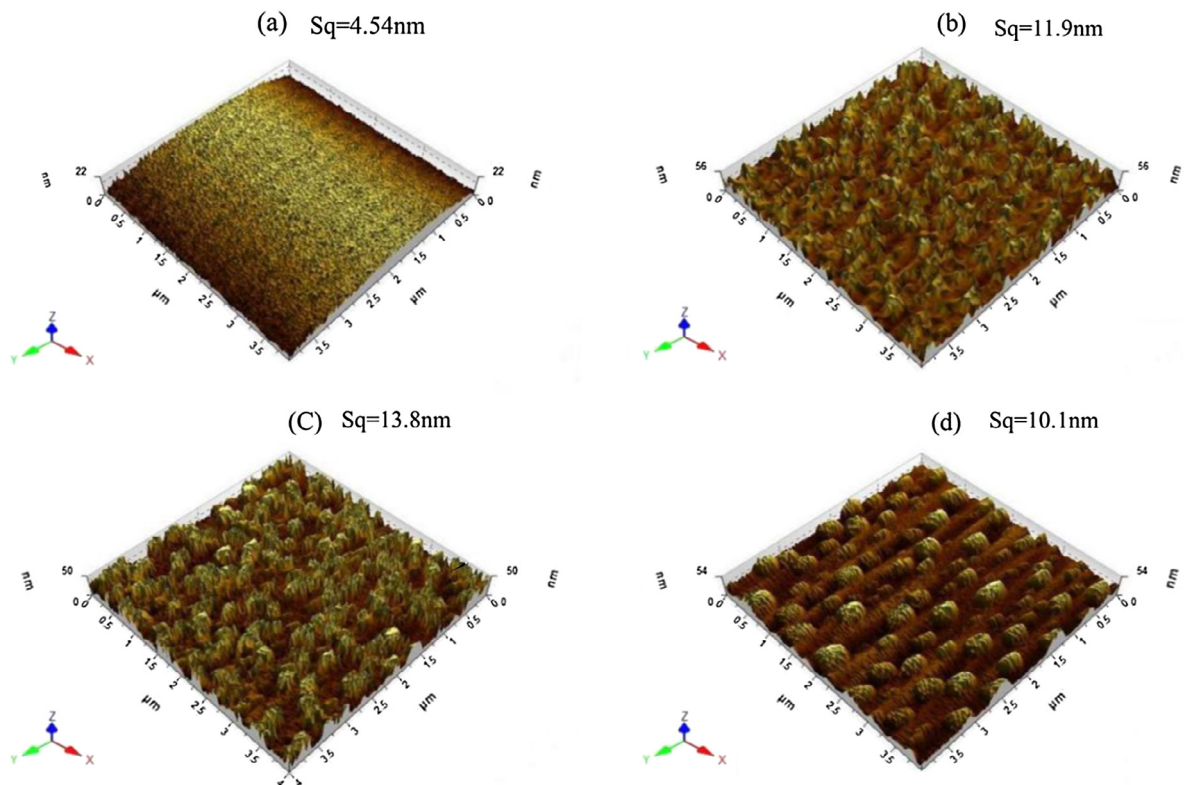


Fig. 4. XPS survey spectra for the surface of S-P-D-3 coating before and after immersed in water for 7 days.

Table 3
Elemental atomic percentages of S-P-D-3 coating before and after immersed in water for 7 days.

Samples	Elemental mole percent (%)				
	C	O	Si	N	S
S-P-D-3 (before)	57.1	25.57	7.76	4.48	5.09
S-P-D-3 (after)	66.38	26.22	2.67	3.62	1.12

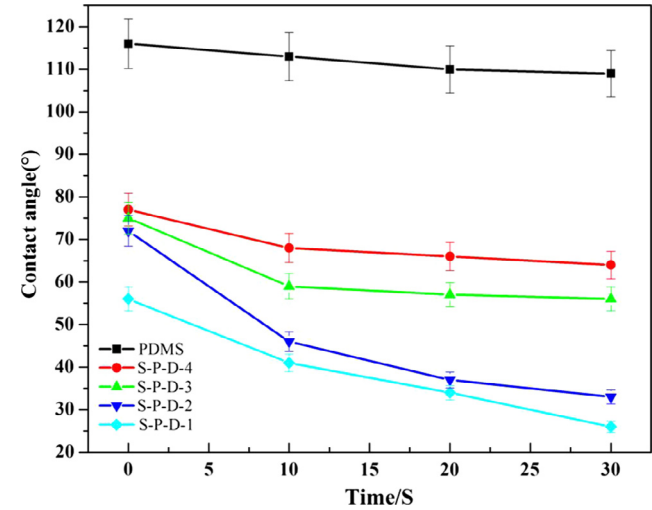


Fig. 5. 3D surface topography of the PDMS-based amphiphilic coatings by AFM observation (a, S-P-D-1; b, S-P-D-2; c, S-P-D-3; d, S-P-D-4).

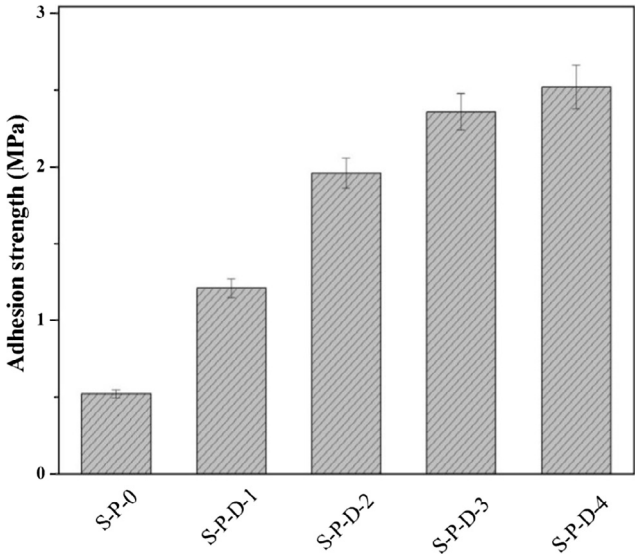


Fig. 6. Water contact angles of the PDMS-based amphiphilic coatings.

Table 4
Cross-hatch adhesion analysis of PDMS-based amphiphilic coatings for different substrates.

Samples	Cross hatch adhesion				
	DMA ratio (w %)	PEG ratio(w%)	glass	Al	steel
S-P-0	0	68.6	0B	0B	0B
S-P-D-1	13.2	53.3	2B	2B	3B
S-P-D-2	21.7	44.8	2B	3B	4B
S-P-D-3	38.2	26.2	3B	3B	4B
S-P-D-4	45.2	19.6	3B	4B	5B

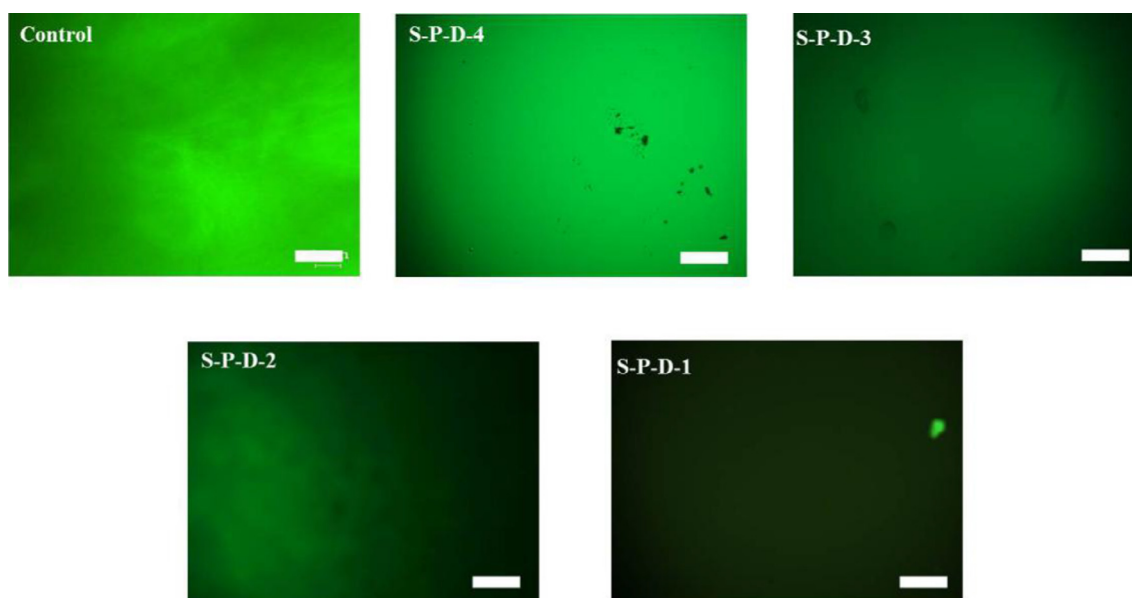


Fig. 7. Pull-off adhesion strength of PDMS-based amphiphilic coatings on steel substrate.

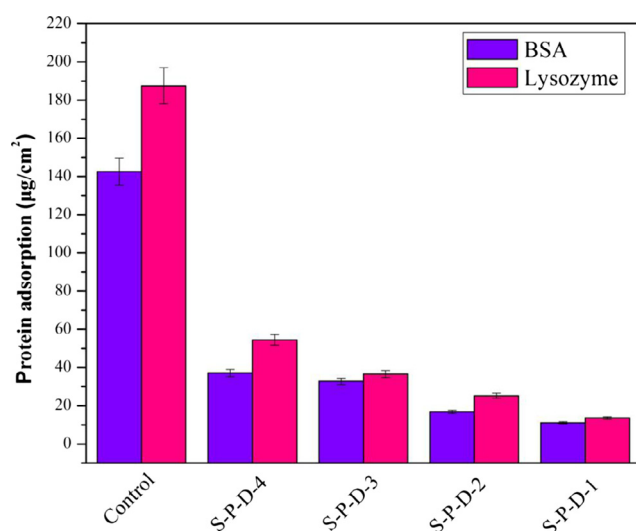


Fig. 8. Fluorescence microscopy images of BSA adsorption on PDMS-based amphiphilic coatings. (The Scale bar :50 μm, Control, PDMS (sylgard 184)).

for the adhesive improvement. Moreover, the hardness and impact resistance performances of coatings are also obviously improved with addition of DMA (See supporting information, Table S1).

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.apsusc.2018.08.214>.

4.7. Nonspecific protein resistance

The amount of protein adsorption on surface is crucial to evaluate the fouling resistance performance of the amphiphilic PDMS-based coatings [23]. Firstly, the qualitative assessment of FITC-labeled BSA adsorbed on the surfaces is characterized by fluorescence microscopy. As shown in Fig. 8, compared with control PDMS coating which shows bright green, all the amphiphilic coatings exhibit very low fluorescence intensity which means a less FITC-labeled BSA adsorbed on coating surfaces, indicating enhanced antifouling properties.

Meanwhile, Fig. 9 shows quantitative BSA and lysozyme protein adsorption on coating surface measured by bicinchoninic acid (BCA) assay. The amount of adsorbed BSA and lysozyme adsorption on

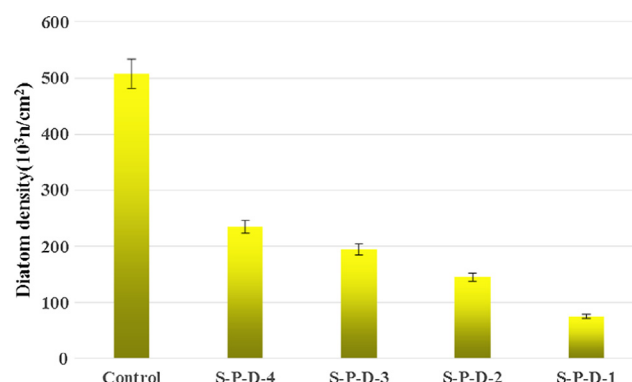


Fig. 9. BSA and lysozyme protein adsorption amounts on PDMS-based amphiphilic coatings. (Control, PDMS (sylgard 184)).

amphiphilic coatings is significantly decreased compared with blank control PDMS coating. Moreover, the increase of PEG content in the coatings will contribute to enhance the antifouling effects. The phenomenon is attributed to two factors. The highly ordered complex surface topography in nanoscale is possibly contributed to the resistance of protein adsorption [44]. The PEG moieties in coatings tend to migrate to surface in water environment, thus generating hydration layer which effectively resists protein adsorption. Interestingly, the adsorbed lysozyme on coating surfaces is higher than that of BSA, which is likely due to the electrostatic interaction between proteins and coating surfaces. The research demonstrates that the PDMS-based amphiphilic coatings exhibit excellent resistance to protein adsorption, which will effectively inhibit the formation of overall biofouling.

4.8. *P. Tricornutum* adhesion assays

In order to evaluate the anti-biofouling performance of the PDMS-based amphiphilic coatings, the *P. tricornutum* as a common fouler in marine is selected as the model. As shown in Fig. 10, the number of diatoms attached to amphiphilic coating surfaces is obviously lower than that on the commercial PDMS coating control ($507 \times 10^3/\text{cm}^2$) after submerging the surfaces in an *P. tricornutum* solution for 7 days. The unique highly ordered heterogeneous surface topography at micro-nano scales on coating surface would provide obstacles for settling diatoms [45]. Moreover, their antifouling property is related to the

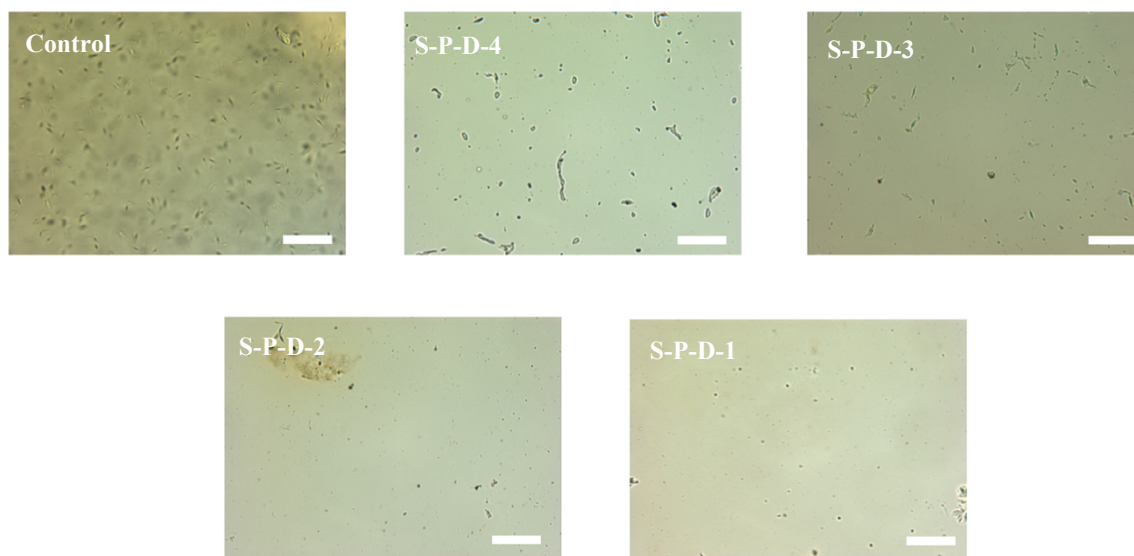


Fig. 10. Average number of *P. tricornutum* cells attached on coating surfaces after 7 days. (Control, PDMS (sylgard 184)).

chemical composition of amphiphilic coatings, the results indicate that the coating containing higher PEG content exhibits better resistance to diatom attachment, due to the formation of hydration layer around PEG segments. Fig. 11 reveals the microscope images of attached diatoms on different coating surfaces. Obviously, the attached diatoms on blank control PDMS coating is more than amphiphilic coatings.

5. Conclusions

In summary, novel biomimetic self-adhesive PDMS-based amphiphilic cross-linked coatings are fabricated via one-step photopolymerization with the combination of three different components, PDMS, PEG and DMA. The resulting PDMS-based amphiphilic coatings exhibit heterogeneous nanoscale surfaces with highly ordered domains,

which can be precisely regulated by varying the compositional ratios of three components. The prepared coatings exhibit good wettability, indispensable adhesion on various substrates, low swelling ratio in water and high thermal stability. The amphiphilic surfaces exhibit significantly improved resistance to nonspecific protein adsorption and *P. tricornutum* diatoms fouling, which is owing to the coexistence between hydrophilic PEG segments and surface morphology. The photopolymerization method offers a simple way to prepare PDMS-based amphiphilic surface with distinctive topography, which may be a direction for the design of anti-biofouling surface. Moreover, the designed biomimetic PDMS-based coatings are eco-friendly, which is suitable for green marine antifouling coatings and potential biomaterial applications.

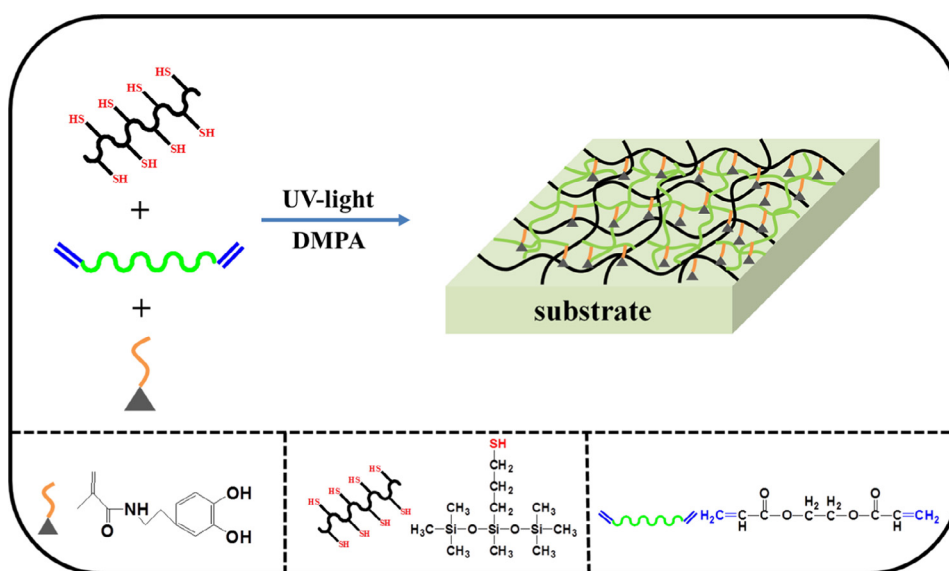


Fig. 11. Microscope images of the attachment of *P. tricornutum* on PDMS-based amphiphilic coatings after 7 days. (The Scale bar :50 μm , Control, PDMS (sylgard 184)).

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